305141
Introduction to Internet of Things

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Course Outine

Part 1: Introduction to Internet of Things

- Overview and Meaning
- Internet of Things Component
- Internet of Things Architecture

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Chapter 01
Introduction

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What is IoT ?

- The Internet of Things (IoT) has become a common news item and marketing trend.
- Beyond the hype, IoT has emerged as an important technology with applications in many fields. IoT has roots in several earlier technologies: pervasive information systems, sensor networks, and embedded computing.
- The term *IoT system* more accurately describes the use of this technology than does *Internet of Things*.
- Most IoT devices are connected together to form purpose-specific systems; they are less frequently used as general-access devices on a worldwide network.

What is IoT ?: *Smart refrigerators Example*

- IoT moves beyond pervasive computing and information systems, which concentrated on data.
- **Smart refrigerators** are one example of pervasive computing devices.
- Several products included built-in PCs and *allowed users to enter information* about the contents of their refrigerator for menu planning.
- Conceptual devices would **automatically scan** the refrigerator contents to take care of data entry.



Refrigerator



PCs, mini PCs



Smart Refrigerator

What is IoT ?: *Smart refrigerators Example*

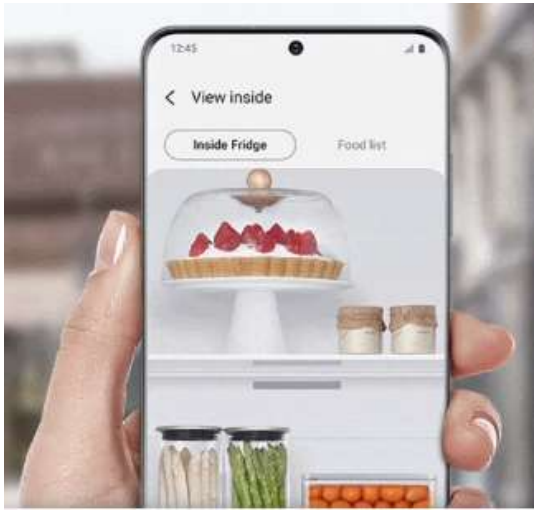
- Samsung Family Hub



Stream your favorite music



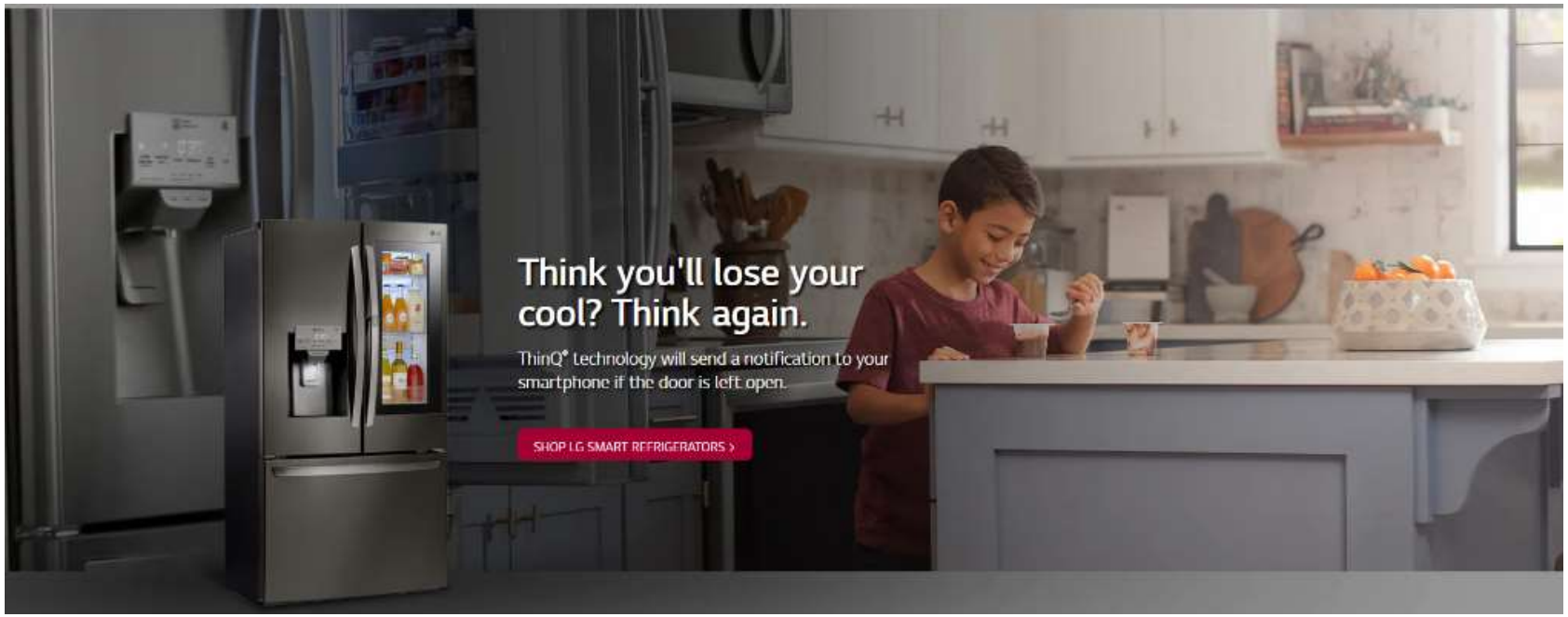
Now with Samsung TV Plus



- https://www.samsung.com/us/explore/family-hub-refrigerator/overview/

What is IoT ?: *Smart refrigerators Example*

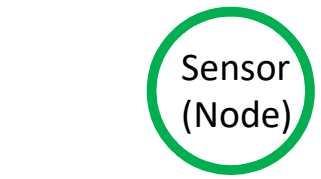
- LG Smart Frigde with ThinQ Application



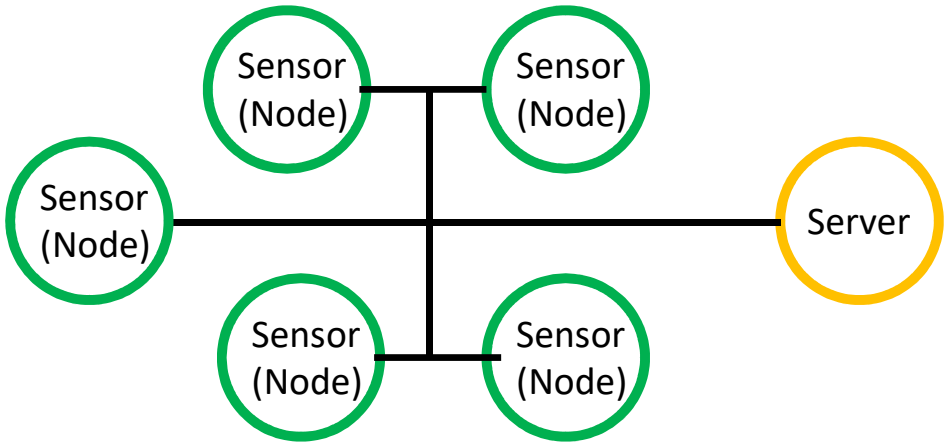
- https://www.lg.com/us/discover/thinq/refrigerators

What is IoT ?: *Sensor Network*

- **Sensor network** research spanned a range of configurations.
- Many of these were designed for **data collection** at *very low data rates*.
- The collected **data** would then be *sent to servers* for **processing**.
- Traditional sensor network research did not emphasize in-network processing.



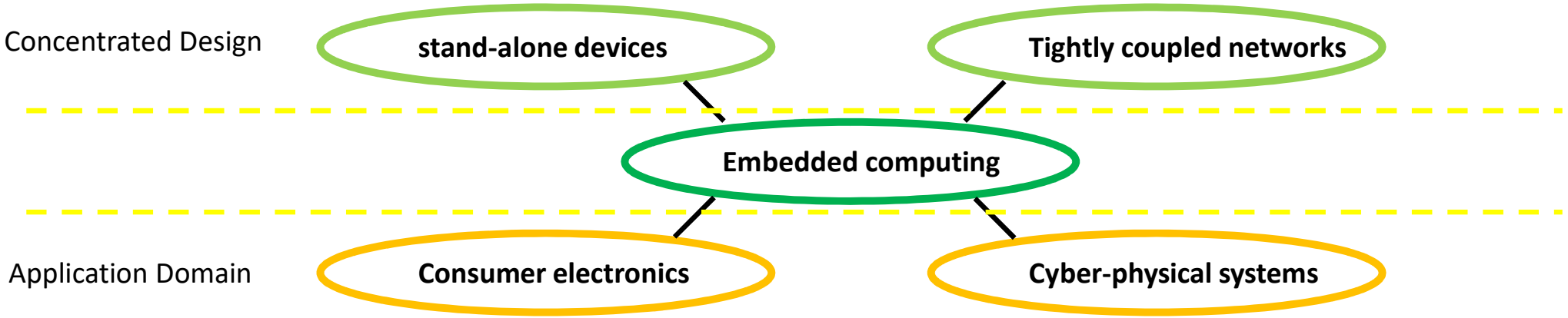
Traditional Sensor Research



IoTl Sensor Research

What is IoT ?: *Embedded Computing*

- **Embedded computing** concentrated on either stand-alone devices or tightly coupled networks such as those used in *vehicles*.
- **Consumer electronics** and **cyber-physical systems** were two major application domains for **embedded computing**; both emphasized engineered systems with well-defined goals.



What is IoT ?: *some definitions*

- Given the wide range of advocates for **IoT** technology, no single, clear definition of the term has emerged.

- We can identify several possibilities:

- **Internet-enabled physical devices**, although many devices don't use the Internet Protocol.
- **Soft real-time sensor networks**.
- **Dynamic and evolving networks** of embedded computing devices

What is IoT ?: *Simple Characteristics*

- We use this term to capture two characteristics.
- First, the system is designed for one or a set of applications, rather than being an agglomeration of Internet-enabled devices.
- Second, the IoT system takes into account the dynamics of physical systems.

- An IoT system may consist primarily of **sensors**; in some cases it may include a significant number of **actuators**.
- In both cases, the goal is to process signals and time-series data.

What is IoT ?: Sensors & Actuators

Sensors



Actuators









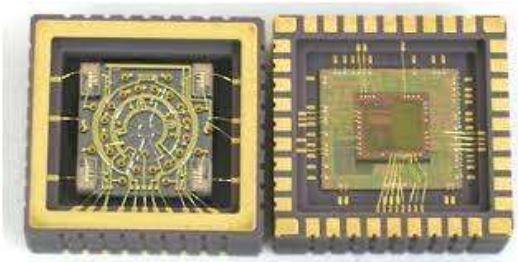


What is IoT ?: *MEMs Sensors*

- Interest in the Internet of Things has been spurred by the availability of microelectromechanical (**MEMs**) sensors.
- Integrated accelerometers, gyroscopes, chemical sensors, and other forms of sensor are now widely available.
- The low cost and power consumption of these sensors enables new applications well beyond those of traditional laboratory or industrial measurement equipment.
- These sensor applications push IoT systems toward signal processing.



accelerometers


gyroscopes


chemical

What is IoT ?: *Power Consumption*

- IoT systems must consume very little power.
- Power consumption is a key factor in total cost of ownership for IoT systems.
- Achieving the necessary power levels requires careful attention to hardware design, software design, and application algorithms.



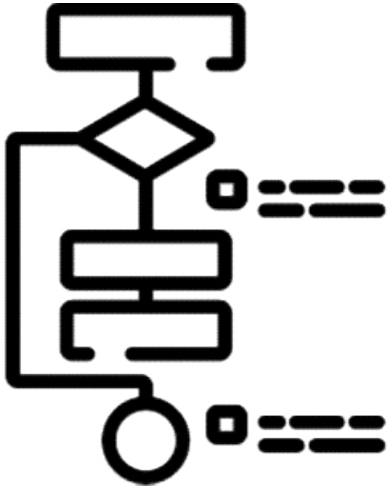
Power consumption



hardware design



software design



algorithms

What is IoT ?: *Security and Safety*

- Security and safety are key design and operational requirements for IoT systems.
- As we have argued elsewhere, safety and security are no longer separable problems.
- The merger of computational and physical systems requires us to merge the previously separate tasks of **safe physical system** design and **secure computer system** design.



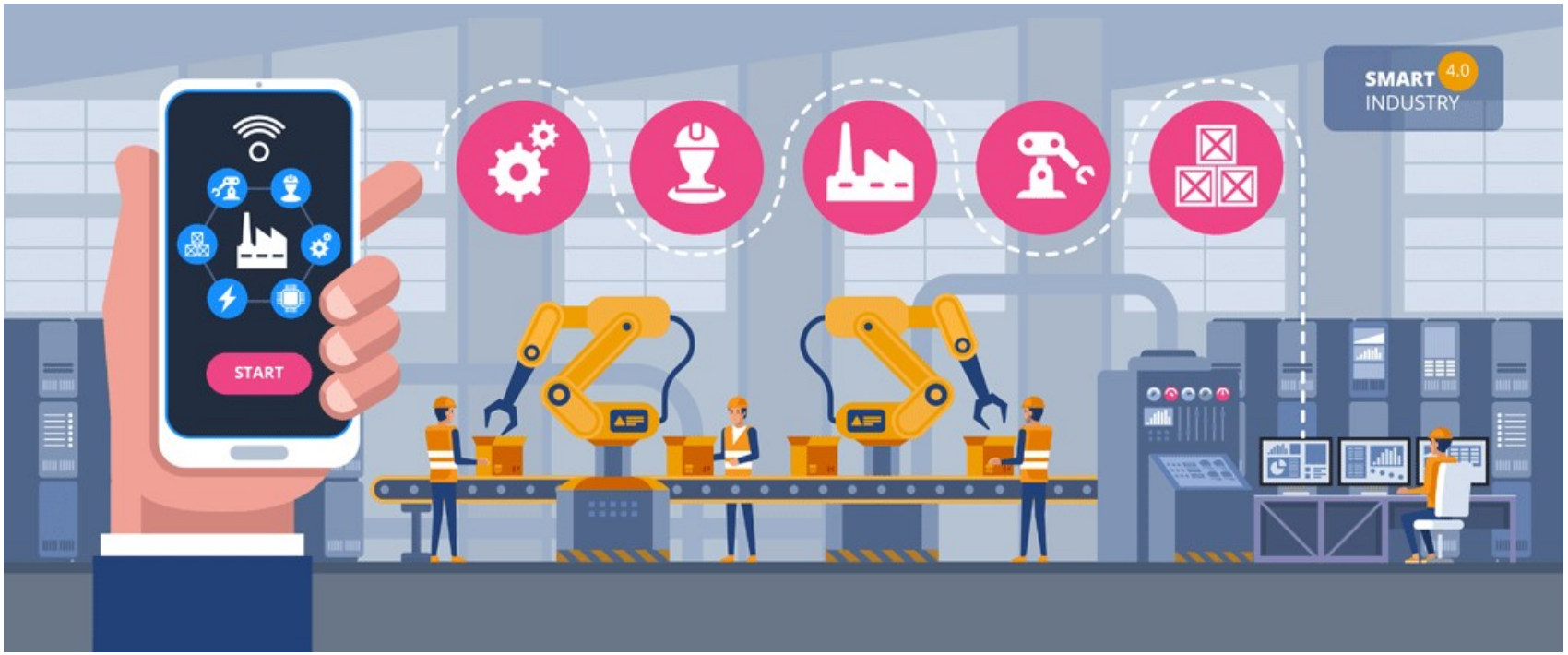
safe physical system



secure computer system

Applications: *Smart Industrial*

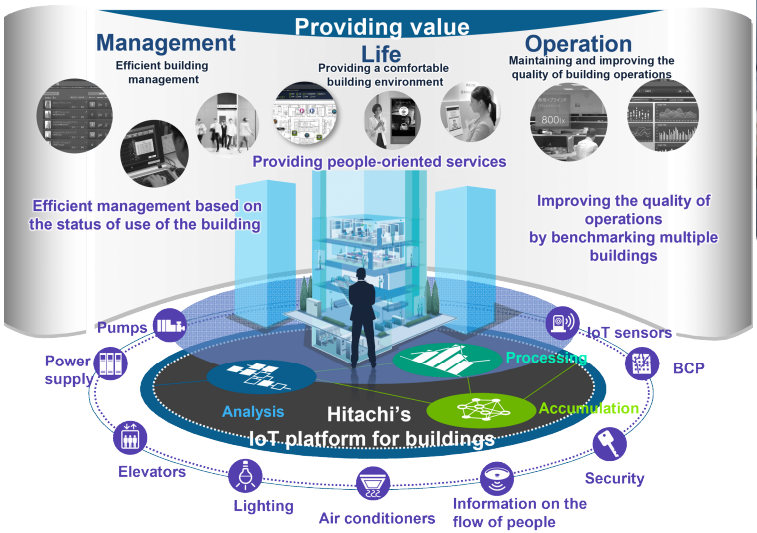
- **Industrial systems** use sensors to monitor both the industrial processes themselves – the quality of the product – and the state of the equipment.



Applications: *Smart Building*

- **Smart buildings** use sensors to identify the locations of people as well as the state of the building.
- That data can be used to control heating/ventilation/air conditioning systems and lighting systems to reduce operating costs.





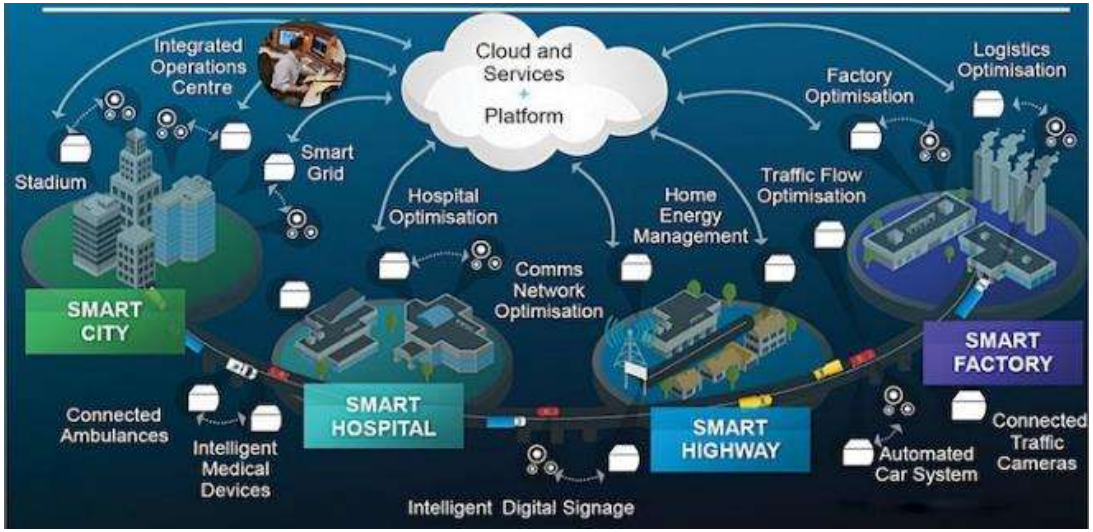




Applications: *Smart Cities*

- **Smart cities** use sensors to monitor pedestrian and vehicular traffic and may integrate data from smart buildings.





Applications: *Vehicles*

- **Vehicles use networked sensors** to monitor the state of the vehicle and provide improved dynamics, reduced fuel consumption, and lower emissions.





Applications: Medical and Health Care

- **Medical systems** connect a wide range of patient monitoring sensors that may be located at the home, in emergency vehicles, the doctor's office, or the hospital.



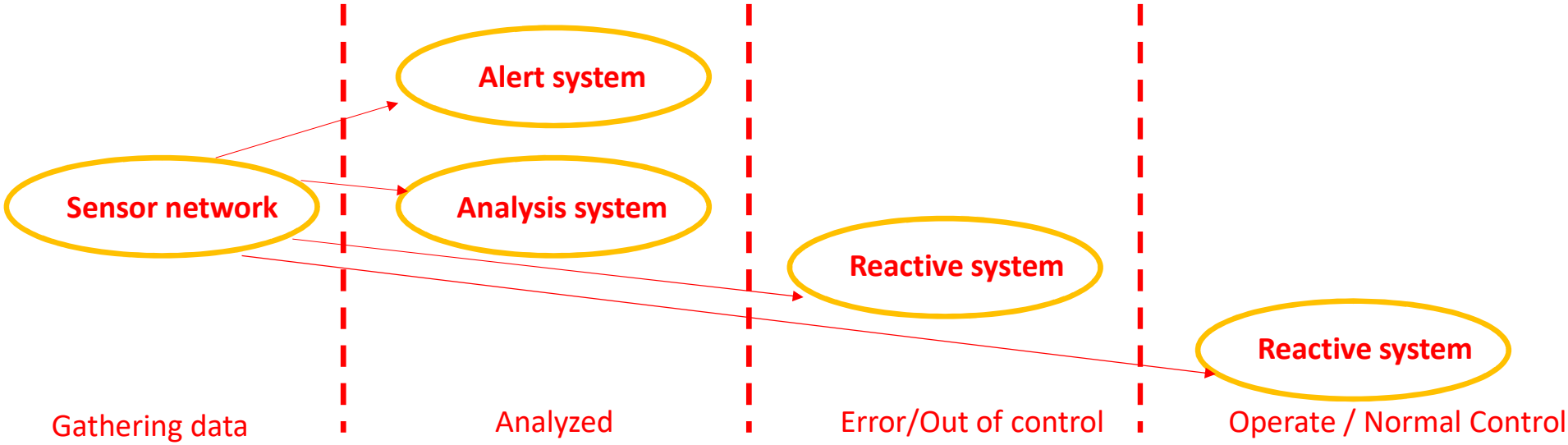


Applications: *Functional Requirements*

- Use cases help us understand the requirements on an IoT system.
- **Sensor network** The system may act strictly as a data gathering system for a set of sensors.
- **Alert system** Data from sensors may be gathered and analyzed. Alerts are generated when particular criteria are met.
- **Analysis system** Data from sensors is gathered and analyzed, but in this case, the analysis is ongoing. Reports on analytic results may be generated periodically – hourly, daily, etc. – or may be continuously updated.

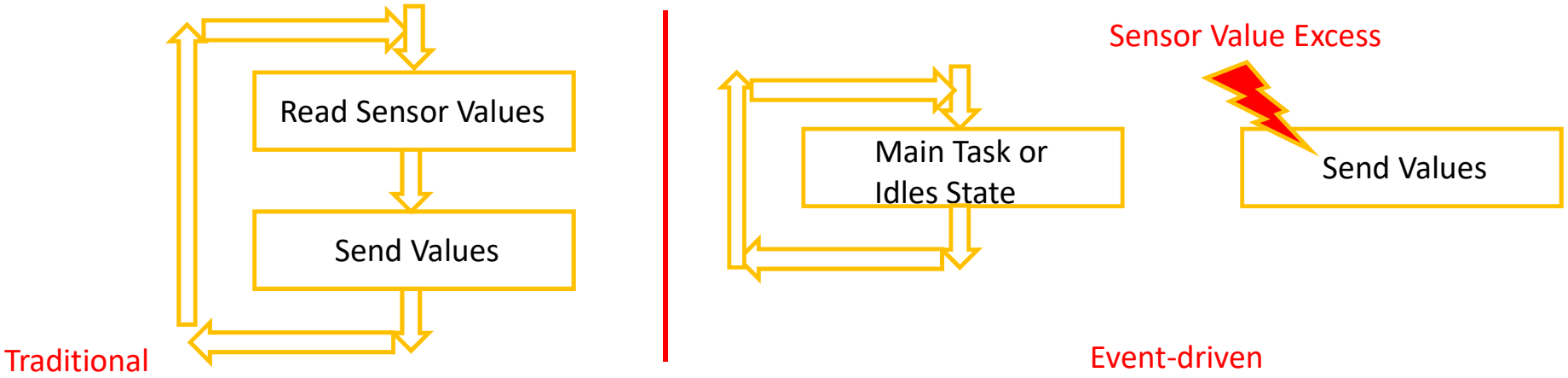
Applications: *Functional Requirements*

- **Reactive system** Analysis of sensor data may cause actuators to be triggered. We reserve the term reactive for systems that don't implement typical control laws.
- **Control system** Sensor data is fed to control algorithms that generate outputs for actuators.



Architectures: *Event-Driven*

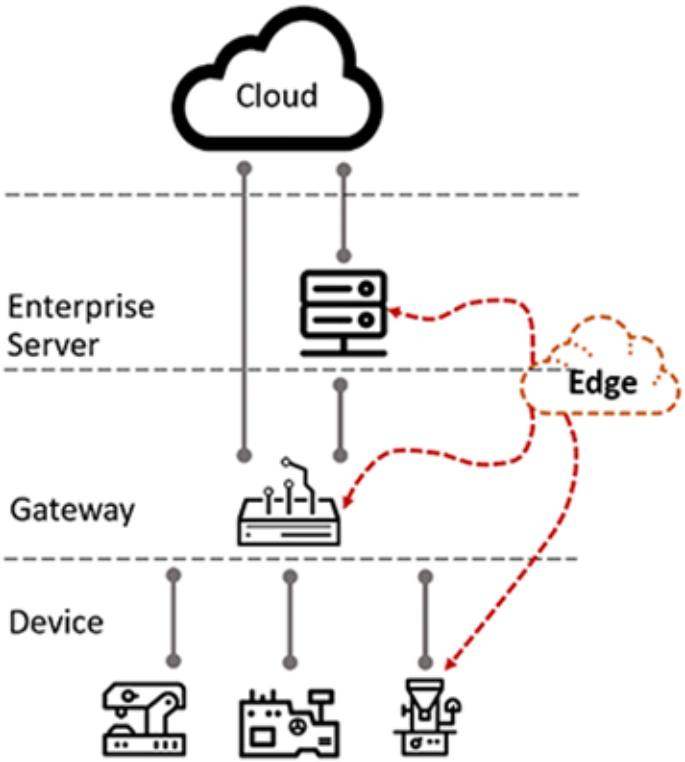
- A key aspect of IoT is ***event-driven*** or ***aperiodic sampling***.
- Traditional digital signal processing and control assume periodic samples resulting in time-series data.
- However, time series consume too much power at the nodes and too much bandwidth on the network.
- Not all applications are amenable to aperiodic data acquisition.



Architectures: *Cloud, Edge, Fog*

- Constraints on power and bandwidth also encourage distributed computing over sensor events.
- Relatively small processors can perform useful processing on many data streams.
- Recognizing interesting events using edge processing reduces the amount of network bandwidth consumed; it also reduces power consumption since wireless communication requires large amounts of power.
- Cloud computing- (centralized servers) or fog computing (servers closer to the edge) can be used to perform further processing on those extracted events.

Architectures: *Cloud, Edge, Fog*



- https://www.nectec.or.th/news/news-public-document/edge-cloud-smartfactory.html

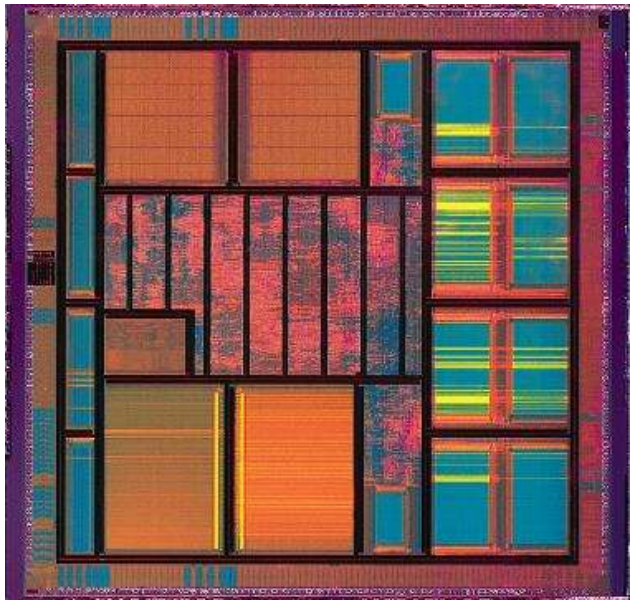
Wireless Networks

- Wireless networks are integral to IoT systems. Wireless network connections simplify installation and operation of wireless networks.
- However, wireless networks introduce some important problems and restrictions.
- Radio communication requires more power than does wired communication.
- Some of the wireless networks used in today's IoT devices were designed for other purposes, such as telephony and multimedia.
- As a result, they are not optimized for event-driven communication and consume significant amounts of power in the communications
- protocol.

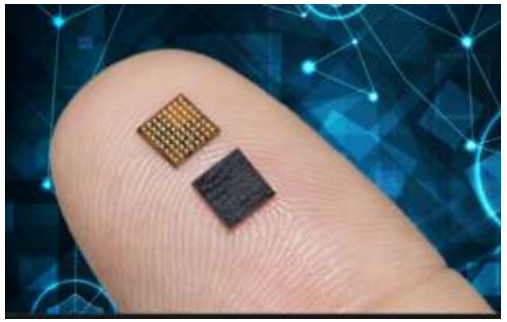
Devices

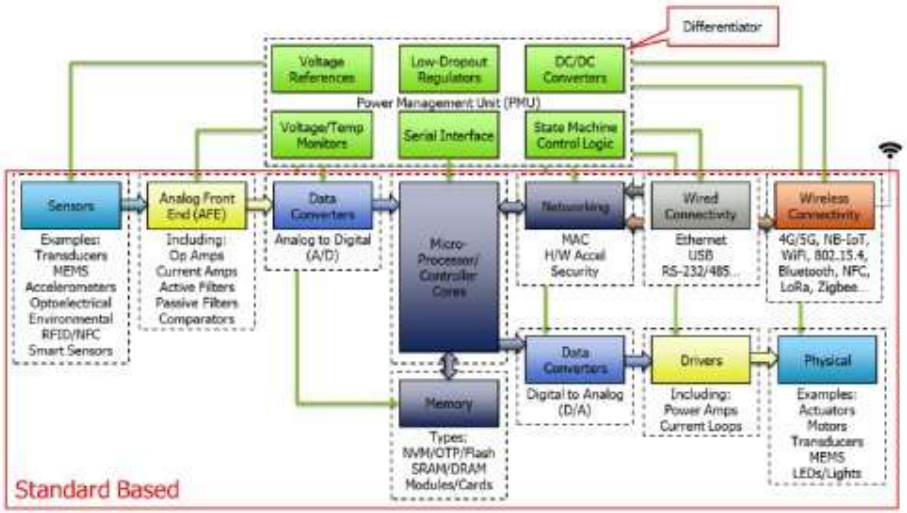
- The characteristics of event-driven systems allow IoT nodes to be relatively simple.
- The realities of low-power operation also push nodes toward relatively low levels of integration.
- **VLSI** technology and Moore's law are key factors in the rise of IoT systems because they allow nodes to be manufactured extremely cheaply.
- Very small chips can provide enough computation, memory, and networking for useful IoT node functions.
- In contrast to traditional microprocessor and consumer electronic applications, where chip areas range around or even higher, chips of several square millimeters are large enough for many IoT node devices.

Devices




VLSI
9050AV 276A6
VGT7737-6126
344S0608-02
APPLE 1988




Differentiator
Voltage References
Low-Dropout Regulators
DC/DC Converters
Power Management Unit (PMU)
Voltage/Temp Monitors
Serial Interface
State Machine Control Logic
Sensors
Examples: Transducers MEMS Accelerometers Optoelectrical Environmental RFID/NFC Smart Sensors
Analog Front End (AFE)
Including: Op Amps Current Amps Active Filters Passive Filters Comparators
Data Converters
Analog to Digital (A/D)
Micro-Processor/ Controller Cores
Networking
MAC H/W Accel Security
Wired Connectivity
Ethernet USB RS-232/485...
Wireless Connectivity
4G/5G, NB-IoT, WiFi, 802.15.4, Bluetooth, NFC, LoRa, Zigbee...
Memory
Types: NVM/OTP/Flash SRAM/DRAM Modules/Cards
Data Converters
Digital to Analog (D/A)
Drivers
Including: Power Amps Current Loops
Physical
Examples: Actuators Motors Transducers MEMS LEDs/Lights
Standard Based

Security and Privacy

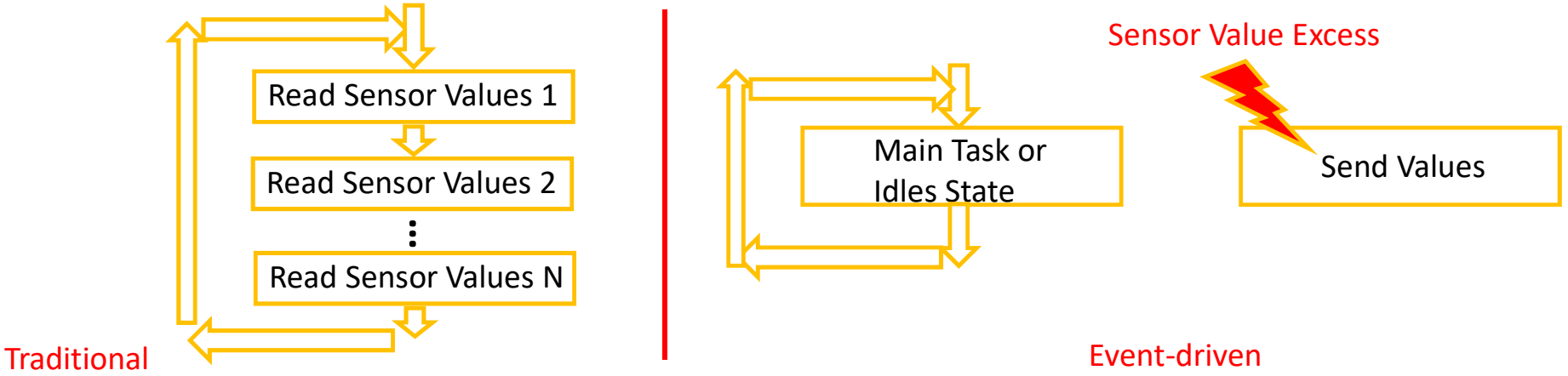
- Security has finally been recognized as an essential requirement for all types of computer systems, including **IoT** systems.
- However, many IoT systems are much less secure than typical Windows/Mac/Linux systems.
- IoT security problems stem from a range of causes: inadequate security features in hardware, poorly designed software with a range of vulnerabilities, default passwords, and other security design errors.
- Insecure IoT nodes create problems for the security of the entire IoT system.
- Because nodes typically have lifetimes of several years, the large installed base of insecure devices will create security problems for some time to come.

Security and Privacy

- **Privacy** is related to security but requires specific measures at the application, network, and device levels.
- Not only must user data be protected from outright theft, but the network needs to be designed so that less-private data cannot easily be used to infer more private data.

Event-Driven Systems

- We believe that the **event** is a fundamental data type in IoT systems and that **event-driven** systems are an **important structuring technique** for IoT.
- Many of the building block technologies used for IoT today show some holdover from traditional, transaction-oriented systems.
- **Event processing** pushes us to treat time as a first-class concept and to consider the relationship between events in event sequences.



Event-Driven Systems

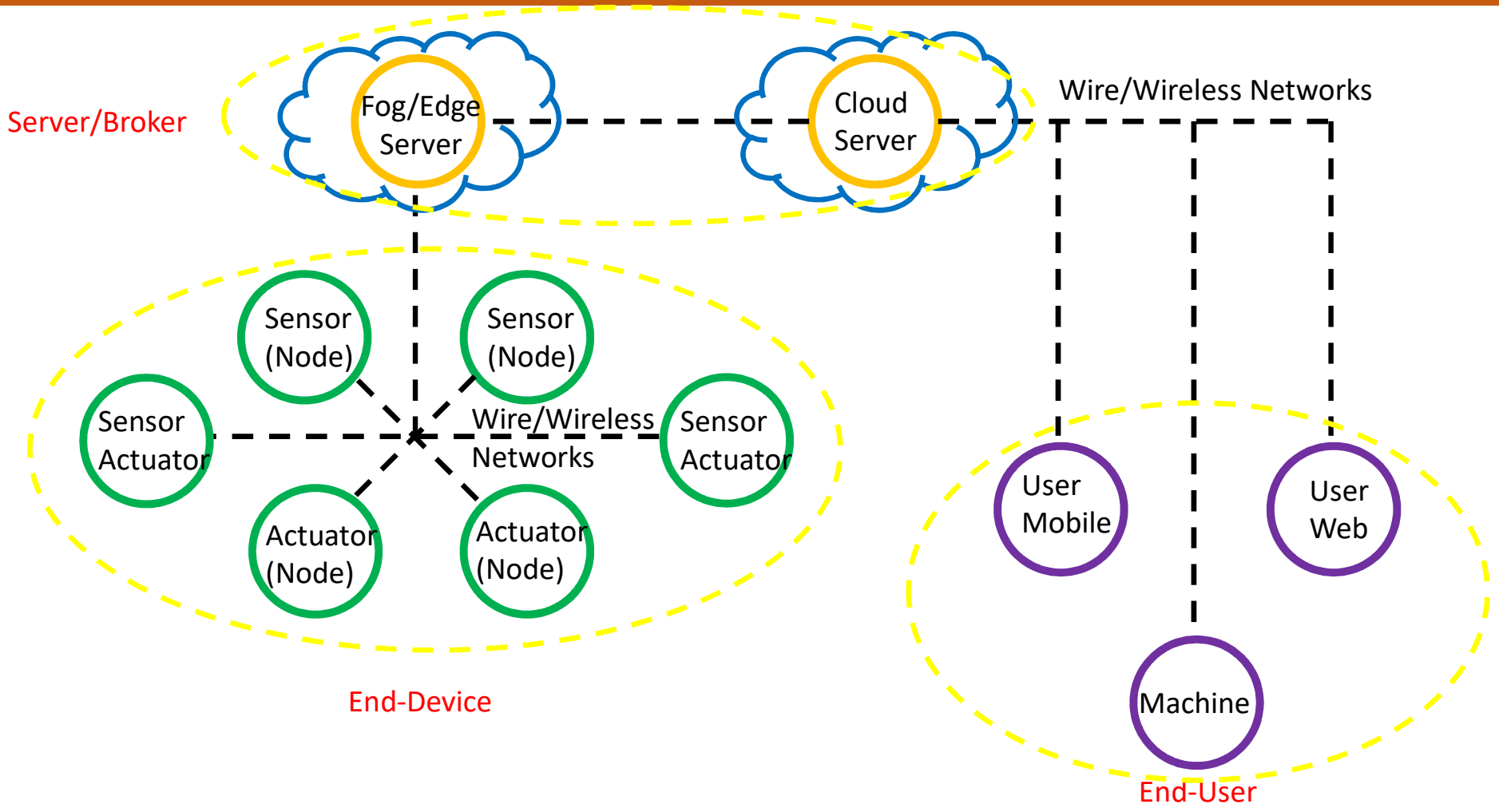
- We use the term ***event*** more broadly then do simulation engineers.
- We consider events as *time-value sets*.
- Event-driven system simulation is widely used for modeling a wide range of engineering systems.
- In that context, an event is generally used to mean a change in the state of a variable.
- Given the decentralized nature of IoT systems, we are willing to consider stuttering – the repetition of an event value – as part of the event model.

Summary: *Meaning*

- The Internet of things (IoT) describes physical objects (or groups of such objects) with sensors, processing ability, software, and other technologies that **connect** and **exchange data** with other devices and systems over the Internet or other communications networks.

https://en.wikipedia.org/wiki/Internet_of_things

Summary: Basic Architecture



Summary: Related Courses and Technologies

